Topic 2: Atoms, elements and compounds - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Atomic structure, isotopes, ions, bonding and structures

Questions in this answer PDF: 53

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 Feb/March Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 28 | Answer source: 0620_m20_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 8: The Periodic Table

Answer points

3(a)

- selenium / Se

3(b)

- Ca has 2 and Cl has outer electrons 7 (1)
- Ca (atoms) lose electrons (1)
- Cl (atoms) gain electrons (1)
- Ca^{2+} (ions) (1)
- Cl^- (ions) (1)

3(c)(i)

- any number in the range 72 - 129°C

3(c)(ii)

- attraction increase (1)
- between molecules (1)

3(c)(iii)

- 3 P - Cl dot cross bonds (1)
- 2 (only) non-bonding electrons to make an octet on P (1)
- 6 (only) non-bonding electrons to make an octet on each Cl (1)

3(d)(i)

- constant concentrations (1)
- rate of forward reaction = rate of reverse reaction (1)

3(d)(ii)

- increased temperature:
- (equilibrium) shifts to LHS (1)
- (forward) reaction is exothermic (1)
- increased pressure:
- (equilibrium) shifts to RHS (1)
- fewer moles (of gas) on RHS (1)

3(d)(iii)

- rate increases and particles have more energy (1)
- more collisions (between particles) occur per second / per unit time
- more (of the) particles / collisions have energy greater than activation energy
- or
- more (of the) particles / collisions have sufficient energy to react
- or
- a greater percentage / proportion / fraction of collisions (of particles) are successful

3(e)

- mol of $LiPF_6$ = 3.04 / 152 = 0.02(00) (1)
- mol of LiF = 0.02(00) x 6 = 0.12(0) (1)
- mass of LiF = 3.12 g (1)

3(f)(i)

- oppositely charged ions
- (ions) are attracted

3(f)(ii)

- any two from:

- physical constants: high boiling point / melting point
- conductivity: conduct (electricity) when aqueous or conduct (electricity) when molten
- solubility: soluble in water

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2020 May/June Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 16 | Answer source: 0620_s20_ms_41.pdf

Answer points

1(a)(i)
- protons
- neutrons

1(a)(ii)
- nucleon number

1(a)(iii)

1(a)(iv)
- 2 : 8 : 8

1(a)(v)
- Al2X3

1(b)(i)
- isotopes

1(b)(ii)
- 12C

1(b)(iii)
- a mole

1(b)(iv)
- Avogadro constant

1(c)
- (3 x 69) + (2 x 71)
-
- 69.8
- =
- =
- Y = Ga / gallium

1(d)(i)
- phosphorus / P

1(d)(ii)
- gains electrons
- three electrons (when forming ion)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_s20_ms_41.pdf
Extra topic(s): Topic 9: Metals

Answer points

2(a)
- metallic (bonding)
- sea of electrons
- positive ions
- attraction between

2(b)(i)
- Mg octet of eight dots
- O octet of six crosses and two dots.
- correct charges on both ions

2(b)(ii)
- 2Mg + O2 -> 2MgO
- species
- balancing

2(c)(i)
- 2Mg(NO3)2 -> 2MgO + 4NO2 + O2
- product species
- balancing

2(c)(ii)
- (thermal) decomposition

2(c)(iii)
- magnesium carbonate
- magnesium hydroxide

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.*

2020 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 13 | Answer source: 0620_s20_ms_42.pdf

Answer points

2(a)(i)
- magnesium 2.8 (all crosses) (1)
- fluorine 2.8 (seven dots and one cross in outer shell) (1)
- Mg2+ and F-(1)

2(a)(ii)
- MgF2

2(a)(iii)
- heat until molten or dissolve in water (1)
- moving ions / mobile ions (1)

2(b)
- two single bonds (1)
- one double bond (1)
- six non-bonding electrons on both F atoms and four non- bonding electrons on O atom to complete the octet in each
- case (1)

2(c)(i)
- forces of attraction between oppositely charged ions / ionic bonds (1)
- strong / need a lot of energy to break / weaken (1)

2(c)(ii)
- forces of attraction between molecules (1)
- weak / need a small of energy to break / weaken (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2020 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 13 | Answer source: 0620_s20_ms_43.pdf

Answer points

3(a)(i)
- same number of electrons
- same electronic configuration

3(a)(ii)
- number of electrons
- number of neutrons
- number of protons
- 3517Cl
- 3717Cl -

3(b)(i)

- displacement / redox

3(b)(ii)

- iodine is less reactive than bromine

3(c)

- magnesium ion has an outer shell with eight crosses
- chloride ion has an outer shell with seven dots and one cross
- chloride has a charge of 1- and magnesium has a charge 2+

3(d)

- energy needed to break bonds = 436 + 243 = 679
- energy released when bonds formed = 2 x 432 = 864
- energy change = 679 - 864 = - AND 185

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.*

2020 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_w20_ms_42.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

1(a)(i)

- (1)
- H (1)

1(a)(ii)

- B

1(a)(iii)

- D

1(a)(iv

- C and G OR C and E

1(b)

- F (1)
- third / outer shell is being filled before second shell is full; second shell has 6 electrons: it should have 8 electrons (1)

1(c)

1(d)(i)

- H-

1(d)(ii)

- aluminium / Al

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2020 Oct/Nov Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 7 | Answer source: 0620_w20_ms_43.pdf

Answer points

2(a)

2(b)

2(c)

- 2,8,8

2(d)(i)

- B, C and E

2(d)(ii)

- A

2(d)(iii)

- D

2(d)(iv)

- B and C

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

2021 Feb/March Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 12 | Answer source: 0620_m21_ms_42.pdf

Answer points

1(a)(i)

- E

1(a)(ii)

- A
- I

1(a)(iii)

- D
- G

1(a)(iv)

- F

1(a)(v)

- H

1(v)(i)

- G and I

1(v)(ii)

- A

1(v)(iii)

- B

1(b)

- same proton number
- different neutron number

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2021 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 5 | Answer source: 0620_s21_ms_41.pdf

Answer points

- Mg: 12 and 13 (1)
- Cu2+: 29 and 27 (1)
- 37(above)
- and17(below) (1)
- Cl (1)
- 1- (1)

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2021 May/June Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_s21_ms_42.pdf

Extra topic(s): Topic 8: The Periodic Table, Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 10: Chemistry of the environment

Answer points

1(a)

- Na or Mg or Al

1(b)

- Mg

1(c)

- S

1(d)

- S

1(e)

- Al

1(f)

- Ar

1(g)

- Si

1(h)

- Al

1(i)

- Cl

1(j)

- Cl

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2021 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 19 | Answer source: 0620_s21_ms_42.pdf

Answer points

2(a)(i)

- isotopes

2(a)(ii)

- mark by row
- 107Ag
- 109Ag+
- p
- n
- e

2(a)(iii)

- average / mean
- 12C

2(a)(iv)

- 50(%)

2(b)

- HNO3

2(c)(i)

- yellow precipitate

2(c)(ii)

- Ag+(aq) + vI-(aq) -> AgI(s)
- AgI (as only product) (1)
- Ag+ and I- (as only reactants) (1)
- state symbols (1)

2(d)

- aluminium (1)
- ammonia (1)

2(e)

- photochemical

2(f)(i)

- any named alkane

2(f)(ii)

- name of organic product derived from 2(f)(i) (1)
- hydrogen chloride / HCl (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.*

2021 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds | Marks: 20 | Answer source: 0620_s21_ms_42.pdf

Extra topic(s): Topic 5: Chemical energetics, Topic 7: Acids, bases and salts

Answer points

5(a)(i)

- 6Li + N2 -> 2Li3N
- N2 as reactant (1)
- rest of equation (1)

5(a)(ii)

- new octet of 8 electrons consisting of 5 crosses and 3 dots in second shell (1)
- charge of 3- (1)

5(b)(i)

- bonds broken [945 + (3 x 160)] = 1425 (1)
- bonds formed (2 x 3 x 300) = 1800 (1)
- energy change = - = 1425 - 1800 = -375 (1)

5(b)(ii)

- Answer must reflect answer in 5(b)(i)
- exothermic
- and
- more energy released (in bond formation) than used/taken in (in bond breaking)

5(b)(iii)

- N with 1 bonding pair with each F (1)
- 2 non-bonding dots for N (1)
- 6 non-bonding crosses for F (1)

5(c)

- ionic bonds in Li3N (1)
- attraction between molecules in NF3 (1)
- weaker attraction (between particles) in NF3 opposite reasoning also allowed (1)

5(d)(i)

- rfm of NH4NO3 = 80 (1)
- mass of N = 2 x 14 = 28 and percentage N = 100 x 28 / 80 = 35% (1)

5(d)(ii)

- fertiliser

5(d)(iii)

- calcium hydroxide

5(e)(i)

- proton acceptor

5(e)(ii)

- 7 < x ■ 11

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2021 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 5 | Answer source: 0620_s21_ms_43.pdf

Answer points

- B: 5 and 6 (1)
- Cl -: 18 and 18 (1)
- 54 and 24 (1)
- Cr (1)
- 3+ (1)
- Question Answer

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2021 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 22 | Answer source: 0620_w21_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

2(a)(i)

- isotopes

2(a)(ii)

- 1 mark for each correct row
- 63Cu
- 65Cu2+
- p
- n
- e

2(a)(iii)

- = (70 x 63) + (30 x 65) or [(4410) + (1950)] or 6360 (1)
- = / 100 = 63.6 (1)
- OR
- = (0.7(0) x 63) + (0.3(0) x 65) or [(44.1(0)) + (19.5(0))] (1)
- = 63.6 (1)

2(b)(i)

- white (1)
- to (light) blue (1)

2(b)(ii)

- 5H2O

2(b)(iii)

- heating

2(b)(iv)

- boiling point (1)
- is 100 °C
- OR
- freezing point (1)
- is 0 °C (1)

2(c)(i)

- blue precipitate

2(c)(ii)

- Alternative suggestion:
- Cu(OH)2 (as only product) (1)
- Cu2+ and 2OH- (as reactants) (1)
- state symbols (1)

2(d)

- 188
- 4.7 / 188 = 0.025(0)
- 0.025(0) x 2 = 0.05(0)
- 0.05(0) x 24.0 = 1.2

2(e)

- 2NaNO3 -> 2NaNO2 + O2
- NaNO2 (1)
- rest of equation (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.*

2021 Oct/Nov Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w21_ms_42.pdf

Answer points

3(a)

- 1 mark for each correct row
- Name
- Relative mass
- Relative charge
- Proton
- +1

- Neutron
- Electron
- 1 / 1840
- -1

3(b)

- Particle
- Number of
- protons
- Number of
- neutrons
- Number of
- electrons
- 3216S
- 3919K+
- 7935Br-
- = row 1 (1) = row 2 (1) =Br (1)
- = 7935 (on left of any symbol) (1) = charge (on any symbol) (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

2021 Oct/Nov Paper 43 Question 4

Main topic: Topic 2: Atoms, elements and compounds | Marks: 6 | Answer source: 0620_w21_ms_43.pdf

Answer points

4(a)

- 1 mark for each row

4(b)

- diamond and graphite

4(c)(i)

- (22 ÷ 44 =) 0.5 (moles) (1)
- 3.01 x 1023(1)

4(c)(ii)

- 1.505 x 1023

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2022 Feb/March Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_m22_ms_42.pdf

Answer points

1(a)

- nitrogen

1(b)

- sodium

1(c)

- iron

1(d)

- silicon

1(e)

- chlorine

1(f)

- cobalt

1(g)

- chlorine

1(h)

- copper

1(i)

- oxygen

1(j)
- zinc

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.

2022 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_s22_ms_41.pdf

Answer points

2(a)(i)
- charge
- relative mass
- -1
- (1)
- (1)
- Mark by column

2(a)(ii)
- 12 (1)
- 12 (1)
- Mark by row

2(a)(iii)
- (they have) 2 more protons than electrons
- OR
- (they have) 2 fewer electrons than protons
- OR
- (they have) 12 protons and 10 electrons

2(b)
- Na+ or Al 3+ (1)
- F- or O2- or N3- (1)
- Ne (1)

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.

2022 May/June Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_s22_ms_42.pdf
Extra topic(s): Topic 8: The Periodic Table

Answer points

3(a)(i)
- atom(s) of the same element
- with different number of neutrons

3(a)(ii)

3(a)(iii)
- 24 protons; 28 neutrons; 21 electrons

3(b)(i)
- blue
- pink

3(b)(ii)
- (act as) catalysts
- variable oxidation state

3(c)(i)
- mobile electrons

3(c)(ii)
- malleability / malleable

3(d)
- high(er) density
- high(er) melting points

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2022 May/June Paper 42 Question 4

Main topic: Topic 2: Atoms, elements and compounds | Marks: 13 | Answer source: 0620_s22_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)

- (pale) yellow and gas

4(b)

- Method 1
- S 25.2 / 32 or 0.78/0.79 …
- and
- F 74.8 / 19 or 3.93 / 3.94…
- ÷ both by 0.7875 = 1 : 5 or SF5
- (254 ÷ 127 = 2 and) S2F10
- Method 2
- 254 -> 25.2/100 and 254 -> 74.8 / 100 OR ‘64' and ‘190'
- 64 / 32 and 190 / 19
- (2 and 10) to give S2F10

4(c)

- N with 1 dot-cross bonding pair with each Cl
- 2 non-bonding dots for N
- 6 non-bonding crosses for Cl

4(d)

- two crosses in first shell of Li
- 7 dots and 1 cross in third shell of Cl
- ‘+' charge on Li on correct answer line and ‘-‘ charge on Cl ion on correct answer line

4(e)

- ionic bonds in LiCl
- attraction between molecules in NCl3
- weaker attraction (between particles) in NCl3 opposite reasoning also allowed

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

2022 May/June Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_s22_ms_43.pdf

Extra topic(s): Topic 9: Metals

Answer points

1(a)

- carbon dioxide

1(b)

- iron(III) oxide

1(c)

- copper

1(d)

- carbon monoxide

1(e)

- glucose

1(f)

- carbon monoxide

1(g)

- copper

1(h)

- oxygen

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*
- *Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.*

2022 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_s22_ms_43.pdf

Answer points

2(a)(i)

- number of protons (are the same) / 16 protons (1)
- number of electrons (are the same) / 16 electrons (1)
- number of neutrons (are different) / 16,17 neutrons (1)

2(a)(ii)

- number of protons is the same as (the number of) electrons

2(a)(iii)

- same number of (outer shell) electrons

2(b)(i)

- (they have) two more electrons than protons

2(b)(ii)

- P3- OR Cl- (1)
- K+ OR Ca2+ (1)

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2022 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 12 | Answer source: 0620_s22_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

3(a)(i)

- covalent

3(a)(ii)

- weak force(s) of attraction between molecules

3(a)(iii)

- no ions
- OR no mobile electrons

3(b)

- 450 °C (1)
- 200 atmospheres (1)
- iron (catalyst) (1)
- N2 + 3H2 ■ 2NH3 (1)

3(c)

- 2NH4Cl + Ca(OH)2 -> 2NH3 + CaCl2 + 2H2O

3(d)

- all 4 NH dot and cross bonds (1)
- single bonding pair between N's and two non-bonding electrons on each N and no non-bonding e on H and nitrogen octet
- complete (1)

3(e)(i)

- proton acceptor

3(e)(ii)

- N2H4 + H2O ■ N2H5+ + OH-

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Oct/Nov Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w22_ms_41.pdf

Extra topic(s): Topic 10: Chemistry of the environment, Topic 8: The Periodic Table, Topic 7: Acids, bases and salts

Answer points

1(a)

- oxygen

1(b)

- carbon

1(c)

- lithium

1(d)

- nitrogen

1(e)

- neon
- 1f
- lithium

1(g)

- boron

1(h)

- nitrogen

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2022 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w22_ms_42.pdf

Answer points

1(a)

1(b)

- giant covalent

1(c)

- silicon dioxide

1(d)

- layers
- hexagon(al) (rings of carbon)

1(e)

- mobile electrons

1(f)

1(g)

- limewater

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.*

2022 Oct/Nov Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w22_ms_43.pdf

Answer points

1(a)

- relative
- charge
- relative
- mass
- +1
- 1 mark for each correct column

1(b)

- 22 (1)
- 17 (1)

- 17 (1)
- 32 and 16 (1)
- S (1)
- 2 - / - 2 / - - (1)

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2023 Feb/March Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 16 | Answer source: 0620_m23_ms_42.pdf
Extra topic(s): Topic 7: Acids, bases and salts

Answer points

2(a)
- metallic

2(b)(i)
- lighted splint and (squeaky) pop

2(b)(ii)

2(b)(iii)
- universal indicator
- February/March 2023

2(b)(iv)
- 2Na(s) + 2H2O(l) -> 2NaOH(aq) + H2(g)
- NaOH as product in equation (1)
- fully correct equation (1)
- state symbols (1)

2(c)(i)
- isotope(s)

2(c)(ii)
- 6Li
- 7Li+
- protons
- neutrons
- electrons
- each row ✔

2(c)(iii)
- (6 -> 10) + (7 -> 90) (= 690) (1)
- 690 / 100 = 6.9 (1)

2(d)
- eight dots in third shell of both K (1)
- six crosses and two dots in second shell of O (1)
- '+' charge on each K on correct answer line and '2-' charge on O ion on correct answer line (1)
- February/March 2023

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_s23_ms_41.pdf

Answer points

2(a)(i)
- N

2(a)(ii)
- B

2(a)(iii)
- F

2(a)(iv)
- C

2(a)(v)

- Li

2(a)(vi)

- Ne

2(b)(i)

- different atoms of the same element with the same number of protons(1)
- different numbers of neutrons(1)

2(b)(ii)

- 10 -> 20 +
- 11 ->80
- (= 1080)(1)
- (1080 ÷ 100 =) 10.8(1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2023 May/June Paper 41 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 9 | Answer source: 0620_s23_ms_41.pdf

Answer points

3(a)(i)

- Na with 2,8 all crosses(1)
- O with 2,8 outer shell with 6 dots and 2 crosses(1)
- + AND 2■(1)

3(a)(ii)

- Na2O

3(b)

- both bonds with 2 dots and 2 crosses(1)
- 2 lone pairs
- (all dots or all crosses) on both oxygen atoms completing all 3 octets(1)

3(c)(i)

- positive ions and negative ions (1)
- strong attraction / strong bonds (1)

3(c)(ii)

- intermolecular forces

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

2023 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 18 | Answer source: 0620_s23_ms_42.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

2(a)

- halogen(s)

2(b)

- same number of outer shell electrons

2(c)

- gas
- pale yellow-green
- gas
- red-brown
- both gases

2(d)(i)

- nucleon number / mass number

2(d)(ii)

- 79Br
- 81Br-
- protons

- neutrons
- electrons
- Each row ✔

2(d)(iii)

- 79 -> 55(%) + 81 -> 45(%)
- 7990 / 100 = 79.9

2(e)(i)

- Cl2 + 2KBr ■ 2KCl + Br2
- KCl as product
- correct equation

2(e)(ii)

- chlorine less reactive than fluorine

2(f)(i)

- white precipitate

2(f)(ii)

- Ag+(aq) + Cl -(aq) ■ AgCl(s)
- AgCl as only product
- Ag+ + Cl - as only reactants (in 1 : 1 ratio)
- state symbols

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.

2023 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 11 | Answer source: 0620_s23_ms_43.pdf

Answer points

2(a)(i)

- Al

2(a)(ii)

- Ar

2(a)(iii)

- Cl

2(a)(iv)

- Al

2(a)(v)

- S

2(a)(vi)

- Cl

2(b)(i)

- 12C

2(b)(ii)

- 24 -> 85(%) + 26 -> 15(%)
- 2430 / 100 = 24.3

2(c)(i)

2(c)(ii)

- Aluminium / Al

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.

2023 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_s23_ms_43.pdf

Answer points

3(a)(i)

- eight crosses in second shell of Mg
- 7 dots and 1 cross in second shell of F

- '2+' charge on Mg ion on correct answer line
- and '-' charge on F ion on correct answer line

3(a)(ii)

- MgF2

3(a)(iii)

- melting

3(b)

- 4 dot and cross single bonds
- 3 pairs of non-bonding e on each Cl and no non-bonding e on Si

3(c)(i)

- molecule(s)

3(c)(ii)

- covalent bonds
- strong bonds
- and
- giant (covalent) structure

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

2023 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 21 | Answer source: 0620_w23_ms_41.pdf

Extra topic(s): Topic 4: Electrochemistry

Answer points

2(a)

- 5p and 5e (1)
- 6n (1)

2(b)(i)

- 20%

2(b)(ii)

- 0.540 / 10.8 = 0.05(0) mol (1)
- 0.05 -> 6.02 -> 1023 = 3.01 -> 1022 (1)

2(c)(i)

- bauxite

2(c)(ii)

- cryolite (1)
- lowers operating temperature OR improves conductivity (1)

2(c)(iii)

- Al3+ + 3e- -> Al
- Al3+ + 3e- on the left-hand side (1)
- equation correct (1)

2(c)(iv)

- (anodes of) carbon react with oxygen (formed at the anode) (1)
- (form) carbon dioxide (1)

2(d)

- (good) conductor (of electricity) (1)
- low density (1)

2(e)

- aluminium oxide layer (1)
- (oxide layer) is unreactive (1)

2(f)(i)

- 2Al + 3F2 -> 2AlF3
- Al and F2 as reactants (1)
- equation correct (1)

2(f)(ii)

- eight crosses in second shell of Al (1)
- 7 dots and 1 cross in second shell of F (1)
- 3+ charge on Al ion on correct answer line and - charge on F ion on correct answer line (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2023 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w23_ms_42.pdf

Answer points

1(a)
- B

1(b)
- F

1(c)
- D

1(d)
- AND E

1(e)
- G

1(f)
- C

1(g)
- B

1(h)
- F

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2023 Oct/Nov Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_w23_ms_42.pdf

Answer points

2(a)(i)
- 59Co
- 65Cu2+
- protons
- neutrons
- electrons
- one mark for each correct row

2(a)(ii)
- (63 -> 70) + (65 -> 30) = 6360 (1)
- 6360 / 100 = 63.6 (1)

2(b)
- (high) density (1)
- (high) melting point (1)

2(c)(i)
- water molecules (1)
- (the water present) in hydrated crystals (1)

2(c)(ii)
- pink (1)
- CoCl2■6H2O (1)

2(c)(iii)
- white (1)
- blue (1)

2(c)(iv)
- heating (the hydrated copper(II) sulfate)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.

2023 Oct/Nov Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w23_ms_43.pdf

Answer points

2(a)(i)

- A, B, F

2(a)(ii)

- D

2(a)(iii)

- C

2(b)

2(c)

2(d)

- 2,8,8

2(e)

2(f)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

2024 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 5 | Answer source: 0620_s24_ms_41.pdf

Answer points

- : Cl : 17 AND 20 (1)
- Cu+: 29 AND 28 (1)
- 33(above) and 16(below) on left hand side of symbol (1)
- S (1)
- 2- (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

2024 May/June Paper 41 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_s24_ms_41.pdf
Extra topic(s): Topic 4: Electrochemistry

Answer points

3(a)

- 2Na + F2 -> 2NaF(2)
- NaF(1)
- Has to be the only product
- equation completely correct(1)

3(b)(i)

- electrons move
- OR
- electrons are mobile
- OR
- electrons flow

3(b)(ii)

- one shared dot and cross(1)
- 6 non-bonding electrons (either) dots or crosses on each fluorine atom to complete both octets (1)

3(b)(iii)

- liquid(1)
- BOTH melting point is below -200 °C AND boiling point is above -200 °C(1)
- OR
- BOTH -200 oC is higher than -220 °C/ melting point AND lower than -188 °C/ boiling point(1)
- OR
- -200 °C is between melting point or -220 °C and boiling point or

- -188 °C(1)

3(b)(iv)

- ionic bonds in NaF(1)
- attraction between molecules or intermolecular forces in F2(1)
- weaker attraction (between particles) in F2 opposite reasoning also allowed (1)

3(c)(i)

- breakdown by (the passage of) electricity(1)
- of an ionic compound in molten or aqueous (state) (1)

3(c)(ii)

- 2H+ + 2e(-) -> H2
- H+ + e(-) on left hand side(1)
- equation fully correct(1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2024 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds | Marks: 17 | Answer source: 0620_s24_ms_42.pdf

Answer points

5(a)(i)

- 32S
- 34S
- protons
- neutrons

5(a)(ii)

- same electronic configuration / structure

5(a)(iii)

- 34 and g

5(a)(iv)

- 1 mole

5(a)(v)

- (32 -> 95) + (34 -> 5) = 3210
- 3210 / 100 = 32.1

5(b)(i)

- eight crosses in second shell of Mg
- 6 dots and 2 cross in third shell of S
- ‘2+' charge on Mg on correct answer line
- and ‘2-' charge on S on correct answer line

5(b)(ii)

- ionic bonds are strong

5(b)(iii)

- mobile ions

5(c)(i)

- H2SO3

5(c)(ii)

- SO32-

5(d)(i)

- the oxidation number of manganese / Mn is +7

5(d)(ii)

- purple
- to
- colourless

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 5 | Answer source: 0620_s24_ms_43.pdf

Answer points

- Cu: 29 and 34
- Cl^- : 17 and 18
- 6430 to left of symbol
- Zn
- 2+

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2024 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_s24_ms_43.pdf
Extra topic(s): Topic 4: Electrochemistry

Answer points

3(a)(i)

- mobile electrons

3(a)(ii)

- double bond of two dots and two crosses
- 4 non-bonding electrons on each oxygen atom

3(a)(iii)

- liquid
- melting point is below -195 °C and boiling point is above -195 °C
- or
- -195 °C is higher than -199 °C / melting point and lower than -191 °C / boiling point
- or
- -195 °C is in between melting point / -199 °C and boiling point / -191 °C

3(a)(iv)

- graphite has a giant covalent structure
- attraction between molecules in carbon monoxide
- weaker attraction (between particles) in carbon monoxide opposite reasoning also allowed

3(b)

- $2K + Cl_2 \rightarrow 2KCl$
- KCl as only product
- correct equation

3(c)(i)

- breakdown by (the passage of) electricity
- of an ionic compound in molten or aqueous (state)

3(c)(ii)

- $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
- any negatively charged OH species losing electrons
- correct ionic half-equation

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2024 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_w24_ms_41.pdf

Answer points

2(a)

- total number of protons and neutrons in the nucleus of an atom

2(b)

- they are the same

2(c)

- 2,8,4
- 2,8,8

2(d)

- 12n + 11e

- 10e
- Ga
- 69 above 31 and to the left of the symbol
- charge of 3 +

2(e)(i)

- atom(s) of the same element
- with different number of neutrons

2(e)(ii)

- (3 -> 203) AND (7 -> 205) (= 2044)
- 204.4

2(e)(iii)

- same number of electrons / same electronic configuration

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2024 Oct/Nov Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds | Marks: 16 | Answer source: 0620_w24_ms_43.pdf
Extra topic(s): Topic 7: Acids, bases and salts, Topic 3: Stoichiometry

Answer points

3(a)

- 78.8 (%)

3(b)(i)

- reacts with acids and with bases to produce a salt and water

3(b)(ii)

- SnO2 + 2NaOH -> Na2SnO3 + H2O (2)
- Na2SnO3
- correct equation

3(c)

- metals form ionic compounds or ionic bonds only
- OR
- covalent compounds contain non-metals only

3(d)(i)

- four single bonds using one dot and one cross
- three pairs of non-bonding electrons on each chlorine and no non-bonding electrons on tin

3(d)(ii)

- attraction between molecules or intermolecular forces in tin(IV) chloride
- tin(II) oxide has a giant ionic structure
- weaker attraction (between particles) in tin(IV) chloride opposite reasoning also allowed

3(e)(i)

- unreactive coating of aluminium oxide

3(e)(ii)

- no reaction
- Mg + Sn2+ -> Mg2+ + Sn

3(f)(i)

- substance that is chemically combined with water
- OR
- containing water of crystallisation

3(f)(ii)

- 2Sn(NO3)2-20H2O -> 2SnO + 4NO2 + O2 + 40H2O (2)
- 2(SnO) + 4(NO2)
- 40H2O

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Feb/March Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 16 | Answer source: 0620_m25_ms_42.pdf

Extra topic(s): Topic 4: Electrochemistry

Answer points

2(a)

- (strong) (electrostatic) attraction between ions
- ions have opposite charges

2(b)

- 8 dots in third shell of each K
- 6 crosses and 2 dots in third shell of S
- ‘+' charge on each K ion on correct answer line
- and ‘2-‘ charge on S ion on correct answer line

2(c)(i)

- 3 correctly positioned positive ions next to given negative
- all ions correct

2(c)(ii)

- cation(s)

2(d)(i)

- electrolysis

2(d)(ii)

- 2Br- -> Br2 + 2e-
- any negative Br species losing electron(s)
- correct ionic half equation
- February/March 2025

2(d)(iii)

- hydrogen
- bubbles
- oxygen water
- bubbles

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2025 Feb/March Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds | Marks: 17 | Answer source: 0620_m25_ms_42.pdf
Extra topic(s): Topic 3: Stoichiometry

Answer points

5(a)

- 55Mn
- 42Ca2+
- protons
- neutrons
- electrons
- Each row ✔

5(b)(i)

- (Mn is a) transition element

5(b)(ii)

- variable oxidation state

5(b)(iii)

- MnO

5(c)(i)

- a substance that reduces another substance and is itself oxidised
- ‘a substance' reduces another substance
- is itself oxidised

5(c)(ii)

- 3Mn3O4 + 8Al -> 4Al2O3 + 9Mn
- Al2O3
- correct equation

5(d)(i)

- 50 -> 0.200÷1000 = 0.01(00)
- 0.01÷4 = 0.0025(0) / 2.5(0) -> 10-3

- = 0.0025 -> 24000 = 60(.0)

5(d)(ii)

- (damp) litmus (paper)
- and
- is bleached / goes white
- February/March 2025

5(d)(iii)

- kinetic energy of particles decreases
- frequency of collisions between particles decreases
- lower percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy
- OR
- Fewer of the collisions / particles have energy greater than / equal to activation energy

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2025 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_s25_ms_41.pdf

Answer points

2(a)

-
- charge
- (1)
-
- mass(1)
- -1
- +1
- 1 mark for each correct column

2(b)(i)

- electron
- neutron
- proton
- 1 mark for each correct column

2(b)(ii)

- (41 x 10) + (39 x 90) = 3920 (1)
- (3920 ÷ 100) = 39.2(1)

2(b)(iii)

- aluminium has only 1 isotope which has nucleon number 27

2(c)

- Ar (1)
- P3- or S2- or Cl - (1)

Easy explanation / reminder

- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2025 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 16 | Answer source: 0620_s25_ms_42.pdf

Answer points

2(a)

- pair of electrons
- electron(s) shared between two atoms

2(b)(i)

- the oxidation number of chlorine / Cl
- (oxidation number) is +1

2(b)(ii)

- O with 1 dot-cross bonding pair with each Cl
- 4 non-bonding dots for O

- 6 non-bonding crosses for each Cl

2(b)(iii)

- (forces of) attraction between molecules are broken (during boiling)
- covalent bonds (between Cl and O atoms) are broken (during thermal decomposition)
- weak attraction between molecules and strong covalent bonds / bonding between atoms correctly applied

2(b)(iv)

- no ions
- no mobile electrons

2(c)(i)

- graphite

2(c)(ii)

- electrons

2(c)(iii)

- 5 correct Si atoms
- 4 correct O atoms

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2025 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds | Marks: 14 | Answer source: 0620_s25_ms_42.pdf

Extra topic(s): Topic 9: Metals

Answer points

5(a)

- I

5(b)(i)

- lithium / Li

5(b)(ii)

- francium / Fr

5(b)(iii)

- lithium / Li

5(b)(iv)

- potassium / K

5(b)(v)

- potassium / K

5(c)

- 8 dots in second shell of each Na
- 6 crosses and 2 dots in third shell of S
- ‘+' charge on each Na ion on correct answer line and ‘2-‘ charge on S ion on correct answer line

5(d)(i)

- isotopes

5(d)(ii)

- mark by row
- p = 37 and 37
- n = 48 and 50
- e = 37 and 36

5(d)(iii)

- 75(%)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2025 May/June Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 10 | Answer source: 0620_s25_ms_43.pdf

Extra topic(s): Topic 11: Organic chemistry, Topic 9: Metals

Answer points

1(a)

- silicon(IV) oxide

1(b)

- propene

1(c)

- aluminium oxide

1(d)

- graphite

1(e)

- propene
- OR
- ethanol

1(f)

- silicon(IV) oxide
- calcium oxide

1(g)

- propane

1(h)

- ethanol

1(i)

- nitrogen

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.*

2025 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds | Marks: 9 | Answer source: 0620_s25_ms_43.pdf

Answer points

2(a)

- protons and neutrons

2(b)(i)

- electron
- neutron
- proton
- 1 mark for each correct column

2(b)(ii)

- (32 -> 95) + (34 -> 5) = 3210
- 3210 / 100 = 32.1

2(b)(iii)

- 12C

2(c)

- Ne
- Na+ OR Mg2+ OR Al 3+

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 Oct/Nov Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds | Marks: 13 | Answer source: 0620_w25_ms_41.pdf

Answer points

1(a)

- particle
- relative
- mass
- relative
- charge
- proton
- neutron
- 0 / nil
- electron
- - 1

- ✓(1)
- ✓(1)

1(b)(i)
- atoms of the same element that have the same number of protons
- but different numbers of neutrons

1(b)(ii)
- they have the same number of electrons and therefore the same electronic configuration

1(c)
- 22
- 16 and 18
- 50 and 22 and to the left of the symbol
- Ti
- 2+

1(d)
- nucleon number

1(e)
- mol Ar = 2.0 ■ 40 = 0.05 (mol)
- number of atoms = 0.05 -> 6.02 -> 1023 = 3.01 -> 1022

Easy explanation / reminder
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 Oct/Nov Paper 43 Question 1
Main topic: Topic 2: Atoms, elements and compounds | Marks: 9 | Answer source: 0620_w25_ms_43.pdf

Answer points
1(a)
- neutron
- proton
- +1

1(b)
- 18
- 10
- 17 above 8 to the left of the symbol
- O
- 2- / -2 / - -

1(c)(i)
- 12C / carbon-12

1(c)(ii)
- 85 (%)

Easy explanation / reminder
- *Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.*

2025 Oct/Nov Paper 43 Question 2
Main topic: Topic 2: Atoms, elements and compounds | Marks: 8 | Answer source: 0620_w25_ms_43.pdf
Extra topic(s): Topic 1: States of matter

Answer points
2(a)(i)
- B

2(a)(ii)
- F

2(a)(iii)
- D

2(b)(i)
- A
- high melting point AND/OR high boiling point
- poor conductor of electricity when solid AND molten

2(b)(ii)
- C

- only conducts electricity when molten

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.